

北京卫蓝新能源科技股份有限公司

Beijing WeLion New Energy Technology Co., Ltd

混合固液锂电芯 SHP320-35 产品规格书

Specification of Hybrid Solid-liquid Lithium-ion Battery SHP320-35

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电芯型号 Cell Model: PL1187187-35Ah电芯容量 Cell Capacity: 35Ah

Both Parties confirm, cells refer to the project 'Abee_Welion_300cells 35Ah pouch-Contract-24.May.2024' follow this specification. For tests beyond this specification, if problems occur, Welion does not assume responsibility.

Avesta Battery&Energy Engineering

Signature

BeijingWeLionNew EnergyTechnology Co.,Ltd

Signature

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1. 适用范围 Scope

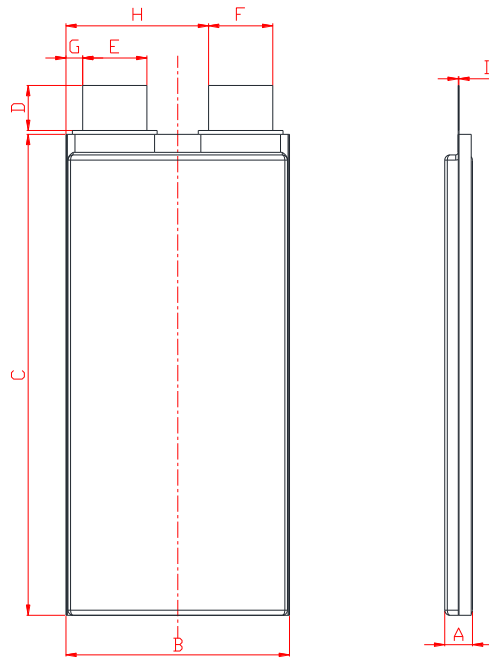
本产品规格书详细描述了北京卫蓝新能源科技股份有限公司（以下简称“卫蓝”）生产的 35 Ah 混合固液锂离子电芯（功率型）新电芯状态的产品性能指标以及产品使用条件。

The purpose of this document is to specify the specifications of 35Ah hybrid solid-liquid lithium-ion (power type) fresh cell for energy storage system (“Product”) to be supplied by Beijing WeLion New Energy Technology Co., Ltd.

2. 产品型号 Cell Model

PL1187187-35Ah

3. 产品尺寸 Product Dimension



项目代码 Code	项目名称 Items	尺寸 Dimension	测试工具 Test tools
A	厚度 Thickness	11±0.3 mm	游标卡尺 Vernier scale
B	宽度 Width	87±1mm	游标卡尺（测试时，折边需紧贴本体） Vernier scale (The edge-fold tightly against the cell body when testing)
C	长度 Length	187±2mm	菲林尺 Film ruler
D	极耳长度（正） Tab length (P)	17±3mm（铝转铜镀镍极耳 Aluminum to copper nickel plated tab）	菲林尺 Film ruler

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		19±3mm (铝极耳 Aluminum tab)	
	极耳长度 (负) Tab length (N)	19±3mm	
E	极耳宽度 (正) Tab width (P)	25±0.5mm	菲林尺 Film ruler
F	极耳宽度 (负) Tab width (N)	25±0.5mm	菲林尺 Film ruler
G	极耳边距 (正) Tab margin (P)	7.2±2.0mm	菲林尺 (测试时, 折边需紧贴本体), Film ruler (The edge-fold tightly against the cell body when testing)
H	极耳边距 (负) Tab margin (N)	56±2.0mm	
I	极耳厚度 Tab thickness	正极耳 Positive tab: 0.3±0.05 mm 负极耳 Negative tab: 0.2±0.05 mm	千分尺 Micrometer

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4. 产品规格 Product Specification

No.	项目 Items		技术指标 Criteria
1	标称容量 Nominal capacity		35Ah (0.33C)
2	标称电压 Nominal voltage		3.65V
3	充电终止电压 Charge cut-off voltage		4.2V
4	放电终止电压 Discharge cut-off voltage		2.75V
5	充电方法 Charge method	超快 Ultrafast	最大充电电流 2C (电芯状态 0%SOC-60%SOC 之间) Maximum charging current 2C (Cell state between 0%SOC-60%SOC)
		快速 Fast	1C (35A) 恒流 (CC) 充电至 4.2V, 再恒压 (CV) 充电至充电电流小于 0.05C (1.75A) 1C (35A) constant current (CC) charge to 4.2V, then constant voltage (CV) charge till charge current decline to $\leq 0.05C$ (1.75A)
		标准 Standard	0.33C (11.55A) 恒流 (CC) 充电至 4.2V, 再恒压 (CV) 充电至充电电流小于 0.05C (1.75A) 0.33C (11.55A) constant current (CC) charge to 4.2V, then constant voltage (CV) charge till charge current decline to $\leq 0.05C$ (1.75A)
6	放电方法 Discharge method	高倍率 High-rate discharge	5C (175A, 60S)
		倍率 Rate discharge	3C (105A)
		标准 Standard	0.33C (11.55 A)
7	循环寿命 (带夹具) Cycle life (with jig)	容量保持率 $\geq 80\%$ Capacity retention $\geq 80\%$	≥ 800 次 ($25^{\circ}\text{C} \pm 2^{\circ}\text{C}$, 0.5C/3C) ≥ 800 cycles ($25^{\circ}\text{C} \pm 2^{\circ}\text{C}$, 0.5C/3C)
8	工作温度 Operating temperature		充电 Charge: $0^{\circ}\text{C} \sim 45^{\circ}\text{C}$ 放电 Discharge: $-20^{\circ}\text{C} \sim 55^{\circ}\text{C}$
9	保护温度 Protection temperature		充电 Charge: $0^{\circ}\text{C} \sim 55^{\circ}\text{C}$ 放电 Discharge: $-20^{\circ}\text{C} \sim 60^{\circ}\text{C}$

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10	储存温度 Storage temperature	短期存储 (1 个月) Short-term (one month): -20°C~45°C 长期存储 (6 个月) Long-term (six months): -10°C~35°C
11	存储湿度 Storage humidity	≤85%RH
12	交流内阻 AC impedance	1.0±0.4mΩ
13	重量 Weight	410±15g

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5. 电性能测试 Electrical Performance Test

5.1 标准充电 Standard Charge

25±2°C, 0.33C (11.55A) 恒流充电至 4.2V, 然后 4.2V 恒压充电, 截止电流 0.05C (1.75A), 静置 30min。

At 25±2°C, charge the cell at 0.33C (11.55A) constant current till the voltage reaches 4.2V, and then charge at 4.2V constant voltage till the current drops to 0.05C (1.75A), rest 30min.

5.2 标准放电 Standard Discharge

25±2°C, 0.33C (11.55A) 恒流放电至 2.75V, 静置 30min。

At 25±2 °C, discharge the cell at 0.33C (11.55A) constant current till the voltage drops to 2.75V, rest 30min.

5.3 电性能参数 Electrical Performance

No.	项目 Items	性能参数 Performance	测试方法 Test methods
1	容量 Capacity	≥35Ah	按照 5.1、5.2 节进行标准充、放电, 记录放电容量。 Charge and discharge the cell according to standard methods of 5.1 and 5.2, and record the discharge capacity.
2	能量 Energy	≥127.75Wh	按照 5.1、5.2 节进行标准充、放电, 记录放电能量。 Charge and discharge the cell according to standard methods of 5.1 and 5.2, and record the discharge energy.
3	倍率放电 Rate discharge	3C 放电容量 > 0.33C 放电容量×95% 3C discharge capacity > 0.33C discharge capacity × 95%	新电芯状态, 电芯标准充电后, 25±2°C 条件下以 3C 放电至 2.75V。 Fresh cell. After being standard charged, the cell is discharged at 3C constant current till the voltage drops to 2.75V at 25±2°C.
4	倍率充电 Rate charge	2C 充电容量 > 总充电容量×80% 2C charge capacity > Total charge capacity × 80%	新电芯状态, 电芯标准放电后, 25±2°C 条件下以 2C 恒流恒压充电到 4.2V, 截止电流 0.05C 充满电。 Fresh cell. After being standard discharged, the cell is charged at constant current and constant voltage of 2C to 4.2V till the current drops to 0.05C at 25±2°C.
5	循环寿命 Cycle life	容量≥初始循环容量×80% Capacity ≥ Initial Capacity ×80%	环境温度 25±2°C的条件下, 0.5C 恒流恒压充电到 4.2V, 截止电流 0.05C, 搁置 30min 后, 以 3C 恒流放

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				电至 2.75V, 搁置 30min, 重复上述步骤, 直至循环容量衰减到初始循环容量的 80% (带夹具 0.02MPa)。 Under the condition of ambient temperature $25 \pm 2^{\circ}\text{C}$, 0.5 C constant current constant voltage charge the cell to 4.2V till the current drops to 0.05C, followed by a 30-minute rest period, and discharge the cell at 3C till the voltage drops to 2.75V, followed by a 30-minute rest period, repeat the above steps until the cycle capacity decays to 80% of the initial capacity (With jig, preload=0.02MPa).
6	- 20°C低温放电性能 Low temperature discharge at -20°C	容量 $\geq$ 标称容量 $\times 80\%$ Capacity $\geq$ Nominal Capacity $\times 80\%$		新电芯状态, $25 \pm 2^{\circ}\text{C}$条件下, 以 0.33C 标准恒流恒压充电到 4.2V, 截止电流 0.05C, 搁置 10 分钟, 然后以 1C 恒流放电到 2.75V, 搁置 10 分钟。再以 0.33C 标准恒流恒压充电到 4.2V, 截止电流 0.05C 充满电; 电芯在 $-20 \pm 2^{\circ}\text{C}$条件下放置 24h, 以 1C 恒流放电至 2.75V。 Fresh cell. Under the condition of ambient temperature $25 \pm 2^{\circ}\text{C}$, 0.33C standard constant current constant voltage charge the cell to 4.2V till the current drops to 0.05C, followed by a 10-minute rest period, then discharge the cell at 1C till the voltage drops to 2.75V, followed by a 10-minute rest period. Then 0.33C standard constant current constant voltage charge the cell to 4.2V till the current drops to 0.05C. The cell is placed at $-20 \pm 2^{\circ}\text{C}$ for 24h and is discharged at 1C till the voltage drops to 2.75V.
7	25°C荷电保持与恢复 Charge retention and recovery at 25°C	保持容量 $\geq$ 标称容量 $\times 94\%$ 恢复容量 $\geq$ 标称容量 $\times 96\%$ Capacity retention $\geq$ Nominal capacity $\times 94\%$ Recovery capacity $\geq$ Nominal capacity $\times 96\%$		新电芯状态, 电芯标准充电后, $25 \pm 2^{\circ}\text{C}$条件下放置 28 天, 以 0.33C 放电至 2.75V, 测量电芯的剩余容量; 0.33C /0.33C 测量电芯的恢复容量。可循环三次, 当有一次达到标准, 即达到要求。 Fresh cell. After being standard charged, the cell is stored in an oven at $25 \pm 2^{\circ}\text{C}$ for 28 days, record the residual capacity in a standard discharge method at 0.33C. And record the recovery capacity in a standard charge and discharge method at 0.33C. Then calculate the retention rate and recovery rate. Cycles times ≤ 3, and if one of them reaches the standard, the requirements are met.
8	45°C荷电保持与恢复 Charge retention and recovery at 45°C	保持容量 $\geq$ 标称容量 $\times 92\%$ 恢复容量 $\geq$ 标称容量 $\times 94\%$ Capacity retention $\geq$ Nominal capacity $\times 92\%$		新电芯状态, 电芯标准充电后, $45 \pm 2^{\circ}\text{C}$条件下放置 28 天。在 $25 \pm 2^{\circ}\text{C}$条件下, 以 0.33C 放电至 2.75V, 测量电芯的剩余容量; 0.33C /0.33C 测量电芯的恢复容量。

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		Recovery capacity ≥Nominal capacity × 94%	可循环三次，当有一次达到标准，即达到要求。 Fresh cell. After being standard charged, the cell is stored in an oven at 45±2°C for 28 days. Under the condition of ambient temperature 25±2°C, record the discharge capacity in a standard discharge method at 0.33C. And record the residual capacity in a standard charge and discharge method at 0.33C. Then calculate the retention rate and recovery rate. Cycles times ≤ 3, and if one of them reaches the standard, the requirements are met.
9	55°C荷电保持与恢复 Charge retention and recovery at 55°C	保持容量≥标称容量×94% 恢复容量≥标称容量×96% Capacity retention ≥Nominal capacity × 94% Recovery capacity ≥Nominal capacity × 96%	新电芯状态，电芯标准充电后，55±2°C 条件下放置 7 天。在 25±2°C 条件下，以 0.33C 放电至 2.75V，测量电芯的剩余容量；0.33C/0.33C 测量电芯的恢复容量。可循环三次，当有一次达到标准，即达到要求。 Fresh cell. After being standard charged, the cell is stored in an oven at 55±2°C for 7 days. Under the condition of ambient temperature 25±2°C, record the residual capacity in a standard discharge method at 0.33C. And record the recovery capacity in a standard charge and discharge method at 0.33C. Then calculate the retention rate and recovery rate. Cycles times ≤ 3, and if one of them reaches the standard, the requirements are met.
10	贮存性能 Storage performance	贮存 3 个月 贮存 6 个月 贮存 12 个月 3 months 6 months 12 months	测量电芯初始容量，电芯标准充电后，记录贮存前的初始状态，分别在室温贮存 3 个月、6 个月、12 个月，测量电芯的最终状态，然后 0.33C/0.33C 循环 3 次记录电芯放电容量。 Record initial status and initial capacity of the cell. Standard charge the cell and store it for 3 months, 6 months and 12 months respectively. Record the final state of cell, then charge and discharge the cell in a standard charge and discharge method at 0.33C for 3 cycles, and record the discharge capacity.
11	过充电 Overcharge	不起火、不爆炸 No fire No explosion	电芯以 1C 标准恒流恒压充电至 4.2V，截止电流 0.05C，然后以 1C 恒流充电截止电压 4.62V，观察 1h。 Charge the cell at 1C constant current till the voltage reaches 4.2V, and then charge the cell at 4.2V constant voltage till the current drops to 0.05C. And charge the cell at 1C constant current till the voltage reaches to 4.62V. Then observe for at least 1h.

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12	强制放电 Forced-Discharge	不起火、不爆炸 No fire No explosion	电芯以 1C 标准恒流恒压充电至 4.2V，截止电流 0.05C，然后以 1C 恒流放电 90min 或者电压为 0V，观察 1h。 Charge the cell at 1C constant current till the voltage reaches 4.2V, and then charge the cell at 4.2V constant voltage till the current drops to 0.05C. And discharge the cell at 1C constant current for 90 min or till the voltage reaches 0V. Then observe for at least 1h.
13	短路 Short Circuit	不起火、不爆炸 No fire No explosion	电芯以 1C 标准恒流恒压充电至 4.2V，截止电流 0.05C，将其正负极用线缆（电阻 $80\pm 20\text{m}\Omega$ ）短接 10min，观察 1h。 Charge the cell at 1C constant current till the voltage reaches 4.2V, and then charge the cell at 4.2V constant voltage till the current drops to 0.05C. Connect the positive and negative terminals directly by a $80\pm 20\text{m}\Omega$ wire for 10min, then observe for at least 1h.
14	跌落 Dropping	不起火、不爆炸 No fire No explosion	电芯以 1C 标准恒流恒压充电至 4.2V，截止电流 0.05C，从 1m 高处自由落体跌落于水泥地上，每个面跌落一次（共 6 次试验） Charge the cell at 1C constant current till the voltage reaches 4.2V, and then charge the cell at 4.2V constant voltage till the current drops to 0.05C. Free fall the cell from a height of 1m to the cement ground, one drop on each surface (6 tests in total).
15	加热 Thermal stability	不起火、不爆炸 No fire No explosion	电芯以 1C 标准恒流恒压充电至 4.2V，截止电流 0.05C，放入温箱中以 $5^{\circ}\text{C}/\text{min}$ 的速率升至 130°C 并保持 30min 后停止加热，观察 1 h（带夹具 0.02MPa ）。 Charge the cell at 1C constant current till the voltage reaches 4.2V, and then charge the cell at 4.2V constant voltage till the current drops to 0.05C. Put the cell into an oven and heat from room temperature to 130°C at the rate of $5^{\circ}\text{C}/\text{min}$, then observe for at least 1h. (With jig, preload= 0.02MPa).
16	挤压 Crushing	不起火、不爆炸 No fire No explosion	电芯以 1C 标准恒流恒压充电至 4.2V，截止电流 0.05C，用半圆柱体挤压头以 $5\text{mm}/\text{s}$ 的速度挤压电芯，至电压为 0 或形变量达到 15%后停止挤压，观察 1h

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Charge the cell at 1C constant current till the voltage reaches 4.2V, and then charge the cell at 4.2V constant voltage till the current drops to 0.05C. Squeeze the cell with a semi cylindrical extrusion head at the speed of 5mm/s till the voltage reaches 0V or the deformation reaches 15%, then observe for at least 1h.

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6. 外观检查 Appearance

不允许有任何影响电芯性能的外观缺陷，如划痕、裂纹、泄漏等。

There shall be no such defect as scratch, crack, leakage, which may adversely affect the performance of battery.

7. 标准测试环境 Standard Test Condition

除非特别说明，本规格书中所有测试均在以下环境条件下进行：

Unless otherwise specified, all tests in this specification are performed under the following environmental conditions:

- 温度 Temperature: $25 \pm 2^{\circ}\text{C}$
- 湿度 Humidity: $\leq 85\% \text{RH}$
- 大气压 Atmospheric pressure: 86kPa~106kPa

8. 包装 Package

8.1 电芯的包装应符合防潮防震的要求，详细包装如下：

The cell shall be packed in accordance with the requirements of moisture-proof and shock-proof. The detailed packaging is as follows:

- 单个电芯使用无色塑料托盘存放，每盘4只；
- Cell is stored in a colorless plastic tray, and there are 4 cells in every tray;
- 外包装为纸箱，每箱18盘共72只。
- The outer packing is carton, where 72 pieces cells in 18 trays per carton is.

8.2 包装箱内应装入随同产品提供的文件：

The packing case shall contain the documents provided with the products:

- 装箱单(指一批多箱包装时)； Packing list (refers to a batch of multiple boxes);
- 产品检验报告。 Product inspection report.

8.3 运输过程建议电芯荷电状态低于30%。

It is advisable to maintain the battery cell's state of charge below 30% during transportation.

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混合固液锂电芯 SHP320-35 产品规格书

Specification of Hybrid Solid-liquid Lithium-ion Battery SHP320-35

文件编号 Document number:

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发布日期 Date issued: 2024.06.10

9. 标识 Identification

9.1 单体电芯产品上应有下列标识:

Cell product shall have the following marks:

- 极性符号 +、- Polar symbol: +/-
- 产品条码 (信息包含产品型号、批号、生产日期信息)

Product barcode (information including product model, batch number and date of production)

9.2 包装箱外壁应有下列标志:

Each packing case shall be marked with:

- 产品名称、产品型号、产品等级、数量、QC pass章
- Product name (solid-state lithium ion cell), product model, product grade, quantity, QC pass seal

10. 存储及其他事项 Storage and Others

10.1 长期储存 Long Term Storage

预计将电芯存放 30 天以上的, 应定期 (建议每隔 3 个月) 按照 5.1、5.2 节中的标准充、放电模式做一次充放电, 然后将 SOC 调整为 20%~40%。

When the products are intended to be stored for a prolonged period of time (more than one month), conduct one cycle of charging and discharging according to the standard charging and discharging mode in Section 5.1 & 5.2 and then adjust SOC to 20%~40%SOC。

10.2 其他事项 Others

任何本规格书中未提及的事项, 须经双方协商确定。

Any matters that this specification does not cover should be conferred between the customers and Beijing WeLion New Energy Technology Co., Ltd.

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11. 保质期 Period of Warranty

保质期是从生产日期（喷码批次）开始起 12 个月。在保质期内，卫蓝可对因电芯生产制造过程中产生缺陷的电芯进行免费更换。

The period of warranty is 12 months from the production date (spraying code batch). During the period of warranty, WeLion can replace batteries that have defects during the manufacturing process for free.

12. 产品责任 Responsibility

客户应当确保严格遵守以下与电芯相关的应用条件，未在以下应用条件下使用电芯，或相关应用条件失效时，造成的经济损失，卫蓝不承担相关经济赔偿责任：

Customer shall ensure that the following application conditions in connection with the products are strictly observed. WeLion is not responsible for the economic losses caused by the failure of the cell under the following application conditions or the failure of the relevant application conditions:

12.1 客户端收到到货电芯后，应在 7 天内完成入库检验，具体参考双方协商的检验规范；

12.1 After receiving the delivered batteries, the client should complete the warehousing inspection within 7 days. Refer to the inspection specifications negotiated by both parties for details.

12.2 工作环境温度范围：充电：0°C~45°C；放电 -20°C~55°C

12.2 Operating environment temperature range: charge 0°C~45°C; discharge -20°C~55°C

12.3 出货态电芯短期储存温度范围（1 个月内）：-20°C~45°C

12.3 Shipped cells short term storage temperature range (within 1 month): -20°C~45°C

12.4 相对湿度：≤85%RH

12.4 Relative humidity: ≤85%RH

12.5 系统成组设计需对电芯施加一定的预紧力，新电芯的预紧力范围为 200N~700N，建议的预紧力公差为±100N。

12.5 The group design of the system requires a certain preload to be applied to the cells. The preload range of fresh cell is 200N~700N, and the recommended preload tolerance is ±100N.

12.6 电芯在使用过程中会产生膨胀力，BOL 满电态到 EOL（80%SOH）满电态膨胀率为 7%，EOL 电

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芯满电态可承受的力约为 10200N。客户在产品的设计过程中需要考虑与电芯产品配套使用的系统结构强度的可靠性，客户需征询卫蓝的建议，并与卫蓝达成一致。

12.6 The product will generate expansion force during use. The expansion rate from BOL@ 100%SOC to EOL@ 80%SOC state is 7 %, The expansion force that EOL cells can withstand at 100 %SOC is about 10200 N. Customers need to consider the reliability of the structural strength of the system used in conjunction with the cell products in the product design and expansion process, customers need to consult WeLion's advice and reach an agreement with WeLion.

12.7 客户应配置电芯管理系统，严密监控、管理与保护每个电芯。

12.7 Customer shall procure that each product shall be used under the strict monitor, control and protection by the BMS.

12.8 客户最终采用的电芯管理系统方案应符合规格书里的 12.10 条款中的相关规定。

12.8 The BMS that the customer ultimately adopts should be in accordance with the relevant specifications stated in 12.10 in technical specification.

12.9 客户应保存完整的电芯系统使用期限内运转的监测数据，用作产品质量责任划分的参考。

12.9 Customer shall keep relevant records of the BMS monitoring data throughout the entire service life of each product, including keeping record of number of occurrence of rush charge, which will be used in the determination and judgment of any product warranty and liability claim entitlement.

12.10 电芯管理系统需满足以下最基本的检测和控制要求

12.10 The BMS shall include the following monitoring and control features as a minimum requirement.

No.	参数 Parameter	产品规格 Specification	保护动作 Action
12.10.1	第一级充电保护 First overcharge protection	$\geq 4.20V$ (100%SOC)	电芯的电压达到 4.20V 时，BMS 申请终止充电。 Stop charging when cell voltage reaches 4.20V.
12.10.2	第二级充电保护 Second overcharge protection	$\geq 4.21 V$	当电芯电压达到 4.21V，BMS 强制终止充电。 Stop charging when cell voltage reaches 4.21V.

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12.10.3	第三级充电保护 Third overcharge protection	$\geq 4.22V$	当电芯电压达到 4.22V, BMS 强制终止充电, 且 BMS 应锁定直到技术人员解决问题。 When the cell voltage reaches 4.22V, the BMS is forced to terminate charging, and the BMS should be locked until technicians solve the problem.
12.10.4	第一级放电保护 First over discharge protection	$V_{min1} \leq 2.75V$	当电芯的电压到达 V_{min1} 时, BMS 申请终止放电。 Minimize the discharging current when cell voltage reaches V_{min1} .
12.10.5	第二级放电保护 Second over discharge protection	$V_{min2} \leq 2.74V$	当电芯的电压到达 V_{min2} , BMS 强制终止放电。 Stop discharging when cell voltage reaches V_{min2} .
12.10.6	第三级放电保护 Third over discharge protection	$V_{min3} \leq 2.73V$	当电芯电压低于 V_{min3} 时, BMS 强制终止放电, 应及时以 0.1C 回充至 20%~40%SOC, 且 BMS 应锁定直到技术人员解决问题。 When the cell voltage is less than V_{min3} , the BMS should be charged back to 20%~40%SOC at 0.1C in time, and the BMS should be locked until technicians solve the problem.
12.10.7	短路保护 Short circuit protection	不允许短路 No short circuit allowed	发生短路时, 应由过流保护装置断开电芯电路。 When a short circuit occurs, the cell shall be disconnected by the overcurrent protection device.
12.10.8	过流保护 Over current protection	参考本规格书第4节第5、6条 Refer to Section 4, paragraphs 5 and 6 of this specification	电芯管理系统控制充放电电流在规格范围内。 Control discharge current by BMS to values within specification.
12.10.9	过热保护 Over temperature protection	$T_{max} > 55^{\circ}C$ (充电) $T_{max} > 60^{\circ}C$ (放电) $T_{max} > 55^{\circ}C$ (Charge) $T_{max} > 60^{\circ}C$ (Discharge)	当温度超过本规格书规定时, 终止充电/放电。 Stop charging and discharging when temperature exceeds specification.

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12.10.10	第一级充电容量保护 First overcharging capacity protection	100%SOC	当电芯容量达到 100%SOC 时, BMS 强制终止充电。 Stop charging when cell voltage reaches 100%SOC.
12.10.11	第二级充电容量保护 Second overcharging capacity protection	105%SOC	当电芯容量达到 105%SOC 时, BMS 强制终止充电。 Stop charging when cell voltage reaches 105%SOC.

备注: 以上 No. 12.10.3、12.10.6、12.10.9、12.10.11 为警示条款, 提请客户注意: 当电芯达到上述任何一项条款描述的指标和参数状态时, 意味着电芯已超出本规格书规定的使用条件, 客户需依“保护动作”及本规格书其他相关规定对电芯采取保护措施, 同时, 卫蓝声明对上述使用状态的电芯质量不承担任何保证责任。

Note: the above No.12.10.3, 12.10.6, 12.10.9, 12.10.11 are the warning clauses, draw the attention of customers: when the battery reaches any of the terms described in the above, it means that the battery has been used beyond the specification. The customer shall take protective measures on the battery in accordance with the protection action and other relevant provisions of this specification. At the same time, WeLion shall not take any responsibility for the quality of the cells mentioned above.

12.11 避免电芯到达过放状态。电芯电压低于 $V_{\min3}$ 时, 电芯内部可能会遭到永久性的损坏, 此时卫蓝的产品质量保证责任失效。根据本规格书第 4 节, 当实际放电终止电压低于标准放电终止电压时, 系统内部能耗降低到最小, 并在重新充电之前延长休眠时间。客户需要培训使用者在最短的时间内重新充电, 防止电芯进入二级放电保护状态。

12.11 Avoid overdischarge of the cell. When the cell voltage is lower than $V_{\min3}$, the inside of the cell may be permanently damaged, at which time the product quality assurance responsibility of WeLion is invalid. According to Section 4 of this specification, when the actual discharge cutoff voltage is lower than the standard discharge cutoff voltage, the internal energy consumption of the system is reduced to a minimum and the hibernation time is extended prior to recharging. The customer needs to train the user to recharge in the shortest possible time to prevent the cell from entering the secondary discharge protection state.

12.12 若预计将电芯存放 30 天以上的, 应定期 (建议每隔 3 个月) 按照 5.1、5.2 节中的标准充、放电模式做一次充放电, 然后将 SOC 调整为 20%~40%。

12.12 When the Products are intended to be stored for a prolonged period of time (more than one month),

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conduct one cycle of charging and discharging according to the standard charging and discharging mode in Section 5.1 & 5.2, and then adjust SOC to 20%~40%SOC。

12.13 电芯避免在本规格书禁止的低温条件下充电（包括标准充电，快充），否则可能出现意外的容量降低现象。电芯管理系统应依照最小的充电温度进行控制。禁止在低于本规格书规定的温度条件下充电。

12.13 Batteries should avoid charging at low temperatures prohibited by this Technical Specification (including standard charging, fast charging), otherwise accidental capacity reduction may occur. Battery management system should be controlled according to the minimum charging temperature. It is forbidden to charge under the temperature stipulated in this technical specification.

12.14 电箱设计中应充分考虑电芯的散热问题，避免由于箱体散热设计问题导致的电芯过热损坏。

12.14 The design of the electric box must fully consider the heat dissipation problem of the cell, to avoid overheating and damage to cells caused by design issues with the electrical box.

12.15 电芯模组设计中应保证电芯完整大面受力均匀，避免影响电芯循环寿命。

12.15 In the design of the battery module, it should be ensured that the whole large surface of the cell is evenly stressed, to avoid affecting the cycle life of cells.

12.16 电箱设计中应充分考虑电芯的防水、防尘问题，电箱必须满足 UL 和 IEC 有关标准规定的防水、防尘等级。避免由于防水、防尘设计缺陷导致电芯损坏（如腐蚀、生锈等）。

12.16 The design of the electric box should fully consider the waterproof and dustproof problems of the battery cell, and the electric box must meet the waterproof and dustproof grades stipulated by UL and IEC. Avoid core damage due to waterproof and dustproof design defects (such as corrosion, rust, etc.).

12.17 卫蓝出货时会根据双方协商的配组要求进行配组，并进行标识和出货。不同标识的电芯不能混用，若有特殊需求可与卫蓝沟通详细解决方案。

12.17 WeLion products are shipped in accordance with integration specifications agreed by both parties, and label the shipment accordingly. Cells with different markings cannot be mixed. Any irregular requirements can be negotiable with WeLion. It is forbidden to mix different P/N batteries in the same battery system, otherwise, WeLion must not be responsible for quality assurance.

12.18 以上供模组/系统设计参考，客户采用的任何模组/系统或其他应用场景的方案设计，都应该做充分的验证并取得相应的第三方认证（如强检），否则出现的模组/系统或其他应用场景的质量问题/损失，卫蓝不承担任何责任。

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12.18 The above is for module/system design reference. Any module/system or other application scenario design adopted by the customer should undergo sufficient verification and obtain corresponding third-party certification (such as mandatory inspection). Otherwise, Welion will not be responsible for any quality problems/losses in the module/system or other application scenario.

13. 文件有效期 Valid Periods

本文件自发布之日起至下次修正日止。

This document is valid since the date of publication until the date of next revision.

14. 保密 Confidentiality

本规格书在没有得到北京卫蓝新能源科技股份有限公司的许可时，不能向第三方泄露，禁止复印或转载，不能对电芯进行拆解。

This specification cannot be leaked to third party, copied and reprinted without the permission of Beijing WeLion New Energy Technology Co., Ltd. All the batteries cannot be disassembled.

15. 发布日期 Publication Date

2024 年 05 月 27 日。

May 27, 2024.

16. 警告事项及注意事项 Warnings and Precautions

16.1 不要将电芯投入火中或加热；

16.1 Do not throw the cell into fire or heating;

16.2 不要将电芯分解拆散；

16.2 Do not disassemble the cell;

16.3 严禁将电芯浸入海水或水中，保存不用时，应放置于阴凉干燥的环境中；

16.3 It is strictly forbidden to immerse the cell in seawater or water. It should be placed in a cool and dry environment when it is stored;

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16.4 禁止将电芯放在热源旁，如火、加热器等；

16.4 Do not place the cell near the heat, such as fire, heaters, etc.;

16.5 充电时请选用锂离子电芯专用充电器；

16.5 Please use a special charger for lithium-ion battery when charging;

16.6 严禁颠倒正负极使用电芯；

16.6 It is strictly forbidden to reverse the positive and negative tabs to use the battery;

16.7 严禁将电芯直接插入电源插座；

16.7 It is strictly forbidden to plug the cell into a power outlet directly;

16.8 禁止用金属直接连接电芯正负极短路；

16.8 It is forbidden to use metal to directly connect the positive and negative tabs of the cell to short circuit;

16.9 禁止将电芯与金属（如发夹、项链等）一起运输或贮存；

16.9 It is forbidden to transport or store batteries together with metals (such as hairpins, necklaces, etc.)

16.10 禁止敲击或抛掷、踩踏电芯等；

16.10 It is forbidden to knock or throw, step on the cell, etc.;

16.11 禁止直接焊接电芯和用钉子或其它利器刺穿电芯；

16.11 It is forbidden to weld or pierce cell with nails or other sharp objects;

16.12 禁止在高温下（炙热的阳光下或很热的汽车中）使用或放置电芯，否则可能会引起电芯过热、起火或功能失效、寿命减短；

16.12 It is forbidden to use or store the battery in high temperature (in the hot sun or in a very hot car), otherwise it may cause the battery to overheat, catch fire or fail to function, and shorten its lifespan;

16.13 禁止在强静电和强磁场的地方使用，否则易破坏电芯安全保护装置，带来不安全的隐患；

16.13 It is forbidden to use it in places with strong static electricity and strong magnetic fields, otherwise it will

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easily damage the safety protection device and bring unsafe hidden dangers;

16.14 如果电芯发生泄露，电解液进入眼睛，请不要揉擦，应用清水冲洗眼睛，并立即送医治疗；

16.14 If the battery leaks and the electrolyte drops into the eyes, do not rub the eyes but wash in time, and go to hospital for treatment immediately;

16.15 如果电芯发出异味、发热、变色、变形或使用、贮存、充电过程中出现任何异常，立即将电芯从装置或充电器中移离并停用；

16.15 If any abnormality such as emitting an odor, emitting heat, discoloration and deformation occur during use, storage or charge, please remove the cell from the device or charger and do not use it anymore;

16.16 废弃电芯应用绝缘纸包住电极以防起火、爆炸；

16.16 The tabs of disused cell should be covered with insulating paper to prevent fire and explosion;

16.17 如果电芯极柱弄脏，使用前应用干布抹净，否则可能会导致接触不良功能失效。

16.17 If the battery tab is dirty, it should be wiped with a dry cloth before use, otherwise it may cause poor contact and function loss.